

APCL

Ambiguity-Preserving Contact Localization

Preserve ambiguity. Move to resolve it.

The idea

A single arm configuration leaves contact location ambiguous along the force line of action. APCL preserves contact hypotheses along the unresolved direction and selects a new orientation to obtain complementary information. It uses joint states and commanded torques, without an object model or joint-torque, force/torque, or tactile measurements.

Available simulation evidence

A paired recovery study on a MuJoCo Franka FR3: 1,200 trials, adaptive stopping at 2–5 views. These figures use the full study before trust-gate filtering.

Method	Median	95th percentile	CBW	95% coverage
CPF / no recovery	9.2 mm	137.6 mm	17.67%	68.5%
APCL	6.6 mm	21.9 mm	0.83%	81.9%

CBW: reported 95% radius < 10 mm and true position error > 20 mm. Empirical coverage is below the nominal 95% level; uncertainty calibration remains incomplete.

What the film shows

Development seed 17, rerun for two views: 166.3 mm error without recovery and 3.2 mm with APCL. Two robots replay their own recorded trajectories: CPF on the left, APCL on the right. Colored diamonds show the weighted contact estimates; gold dots show the true contacts. Each variant selects its own next orientation. 3D positions are to scale, with enlarged markers for visibility. Motion is slowed; estimates update only at saved snapshots.

Scope and limitations

The simulation uses a rigidly grasped payload, one object-fixed contact point, an approximately world-fixed applied force, quasi-static measurement windows, and regulated tool position. It does not model a physical pusher in the replay.

A batch re-estimate on the same measurement windows can match filter accuracy and yield better-calibrated uncertainty. APCL is not established as universally more accurate or faster. Hardware validation is pending. This note is not a submitted or accepted journal article.

Provenance

Source: `claude_try/results/main.jsonl` (1,200 trials).
SHA-256: `f5028492078b81dde049b0368eb7ecc02f`
`2f5e2085f148dcd7c36c8bb13f4a45c`

The companion data archive contains per-trial CSV records, episode trajectories, weighted particle snapshots, metadata, and source hashes.